

Scintillations



“All of the animals except man know that the principle business of life is to enjoy it.”

-Anonymous

Solav
PET CT



**SPECT
LAB**

nuclear medicine services

Visit us at : www.spectlab.com

Volume II / Issue III May - June 2009

Case15. Solitary pulmonary nodule (SPN)

Metabolically Active SPN

Contributed by :

Dr. D Kelkar, Dr. U.Kurlekar, Dr. Jamkhandikar, Dr. Dhanapathi
Dr. Shrikant Solav, Dr. Sujit Joshi ,Dr.Mandolkar.

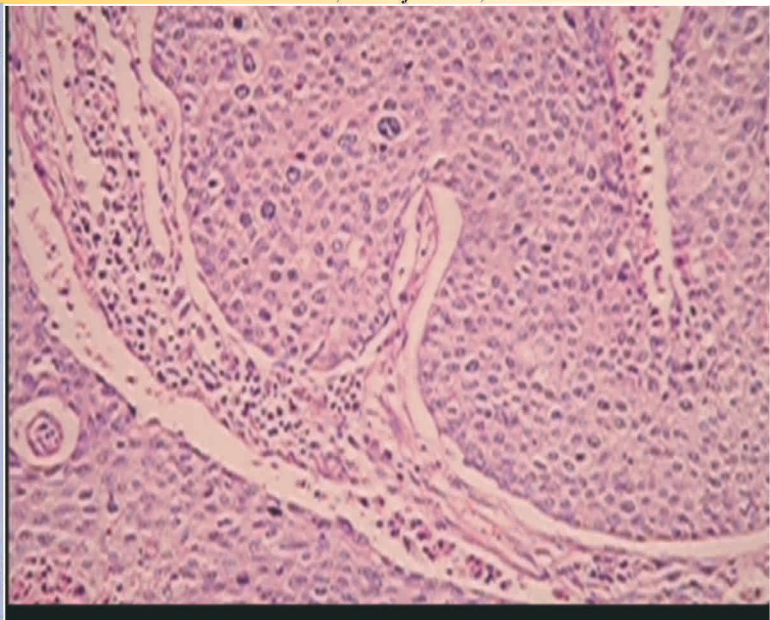
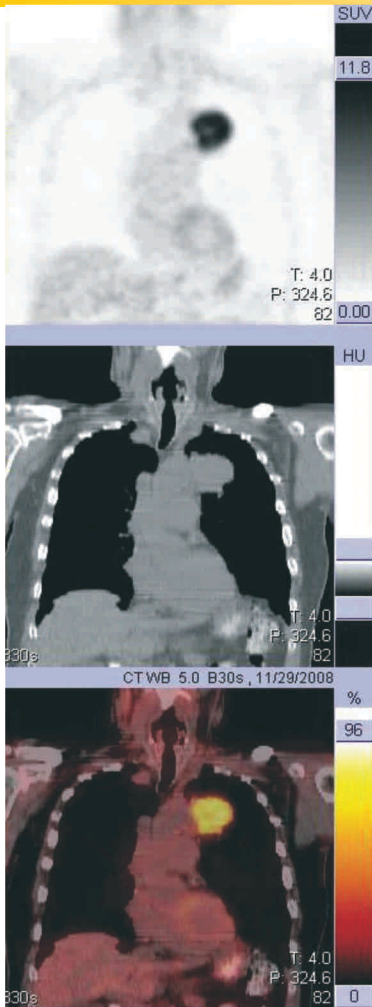


Fig 15B shows sheets of large cells with some appearing bizarre consistent with large cell carcinoma.4

This 73 years old gentleman presented with dry cough. His chest radiograph revealed soft tissue density lesion in the left lung. CT scan confirmed a solitary pulmonary nodule. An FNAC showed scanty cellularity, with no malignant cells seen. There was no evidence of AFB.

PET CT Scan revealed intense metabolic activity in the nodule.(Fig15A) Hence excision biopsy was suggested. The histology confirmed malignant large cell tumor of the lung. (Fig15B))

Fig 15A showing metabolically active left upper lobe pulmonary nodule

Metabolically Inactive SPN

Contributed by :

Dr. Nitin Abhyankar, Dr. Dhanapathi, Dr. Shrikant Solav,
Dr. Aparna Joshi, Dr. Avinash Pradhan

This 50 years old gentleman with longstanding history of Diabetes was incidentally found to have solitary pulmonary nodule on routine radiograph of the chest which was confirmed on CT scan. (Fig 15C)
PET CT revealed ametabolic nodule (cold) in the left lung. (Fig15 D) Such ametabolic nodules are usually benign. However they could be seen in bronchioloalveolar carcinoma or carcinoid tumor, although the incidence of both these conditions is low. Histology revealed inflammatory lesion with no evidence of malignancy.(Fig 15E)

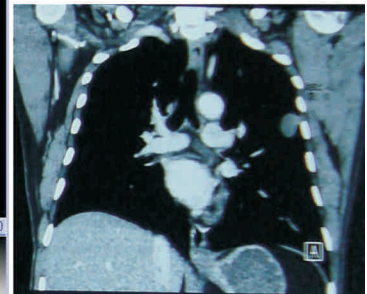
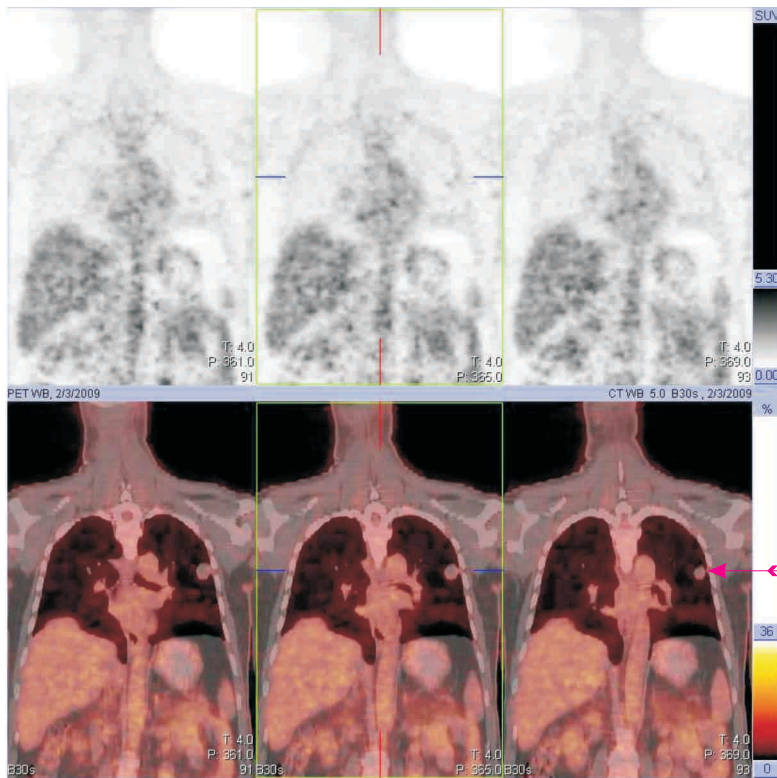


Fig 15C CT chest showing left mid zone pulmonary nodule

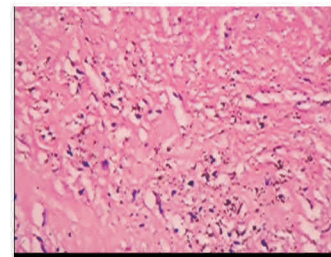


Fig 15 E Histology showing inflammatory lesion.

Fig 15 D PET CT reveals absence of metabolic activity in the nodule

Salient features of Solitary pulmonary nodule:

Features favoring malignancy-

1. Nodules with higher SUV (> 2.5)
2. If the margins are spiculated.
3. 2/3 of primary affect the upper lobes.
4. Metastatic nodules affect the lower lobes.
5. A lesion is called a nodule if less than 3cm in size and it is addressed as a mass when the size is > 3 cm.
6. Likelihood of malignancy is higher if the size is > 2 cm.
7. Doubling time: time taken for the nodule to increase in diameter by 26%.
8. Majority of carcinomas have a doubling time between 30 and 200 days.

Features favoring Benign lesion:

1. Smooth margins
2. Absence of FDG uptake (SUV < 2.5)
3. Doubling time < 30 days.
4. Calcified nodule.

False positive:

Infection, granuloma, hamartoma.

False negative:

Bronchoalveolar carcinoma, carcinoid, hyperglycemia during FDG PET scan.

Case 16. A case of osteomyelitis that mimicked neoplasm

Contributed by :

Dr Yogesh Panchwagh, Dr K H Sancheti, Dr Anand Rahalkar, Dr Dhanpati, Dr Shrikant Solav, Dr Sujit Joshi.



Fig 16 A



Fig 16 A



Fig 16 B

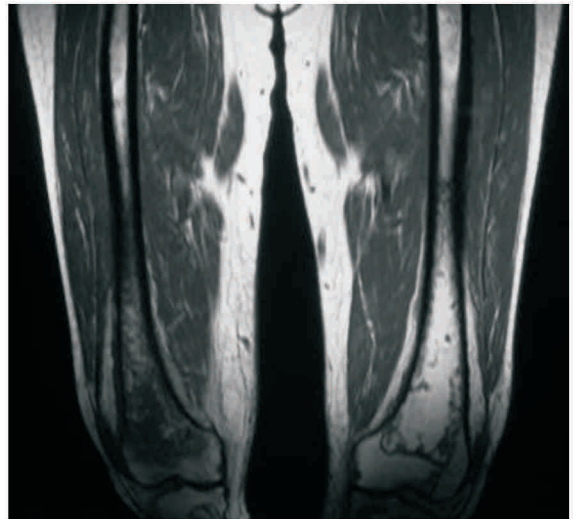


Fig 16 B

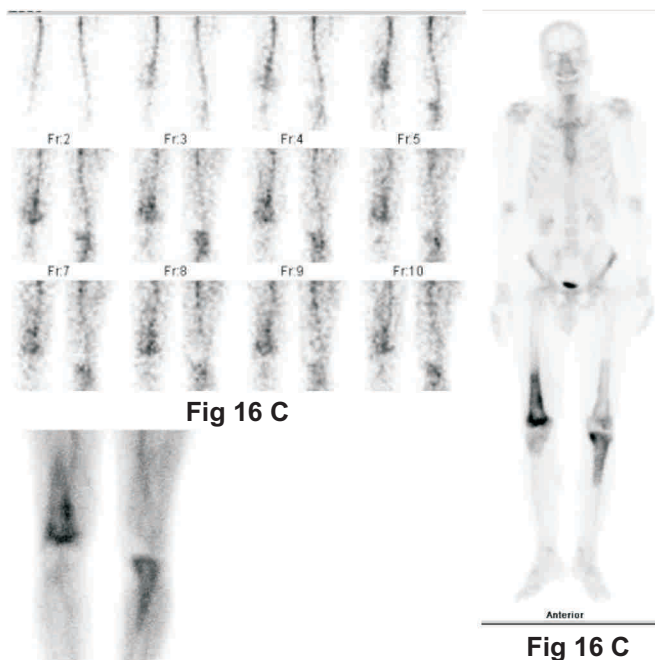


Fig 16 C



Fig 16 C



Fig 16 D

A 39 yrs old gentleman presented with painful swelling in the lower right thigh of 15 days duration. He had a history of fever on and off over the last one month. This otherwise healthy gentleman had neither symptoms at any other site nor any stigmata suggestive of an immuno-compromised status.

A plain radiograph (Fig 16A) of the right femur revealed a lytic and sclerotic lesion in the distal femur with interrupted periosteal reaction. A similar appearing lesion was noticed in the proximal left tibia. Radiological differential diagnosis was either a primary malignant bony neoplasm or a haematolymphoid malignancy. The diagnosis of acute osteomyelitis was considered as rare possibility. M.R.I. scan (Fig 16B) of both femurii revealed lesions bilaterally in the meta-diaphysis. There was patchy involvement with periosteal reaction and soft tissue component in the right femur. 3 phase Tc-99-MDP bone scan (Fig 16C) revealed uptake at both distal femurii and upper end of the left tibia. Given the clinico-radiological profile a possibility of haematolymphoid malignancy was considered most likely.

A J needle biopsy was done for the symptomatic distal femur lesion. At the time of biopsy frank purulent collection drained out from the right femur. (Fig 16 D) This was sent for culture and sensitivity studies. Culture revealed growth of E Coli. Histopathology of the bone revealed osteomyelitic features. A biopsy from the other sites also revealed similar picture. The patient was treated with decompression of the bone with a window in the cortices and intravenous antibiotics.

Infections do sometimes masquerade as a neoplasm. However this case is unique since the infective aetiology affected the metadiaphyseal bone simultaneously at 3 long bone sites in an otherwise healthy adult.

Brief features: Osteomyelitis is a suppurative process causing progressive destruction of bone. Nuclear medicine methods to diagnose osteomyelitis: 3phase bone scan, Technetium-WBC scan, Gallium-67 scan, Sulphur colloid bone marrow scan in correlation with WBC scan- Positive lesion on WBC that is negative on sulphur colloid to be interpreted as positive for infection. FDG PET study shows increased metabolism in affected sites.

Case17. Crossed cerebellar diaschisis and crossed hemispheric diaschisis

Contributed by :

Dr. Shreegiri, Dr. C Raman, Dr. Shreerang Gokhale, Dr. Dhanpathi, Dr. Shrikant Solav.

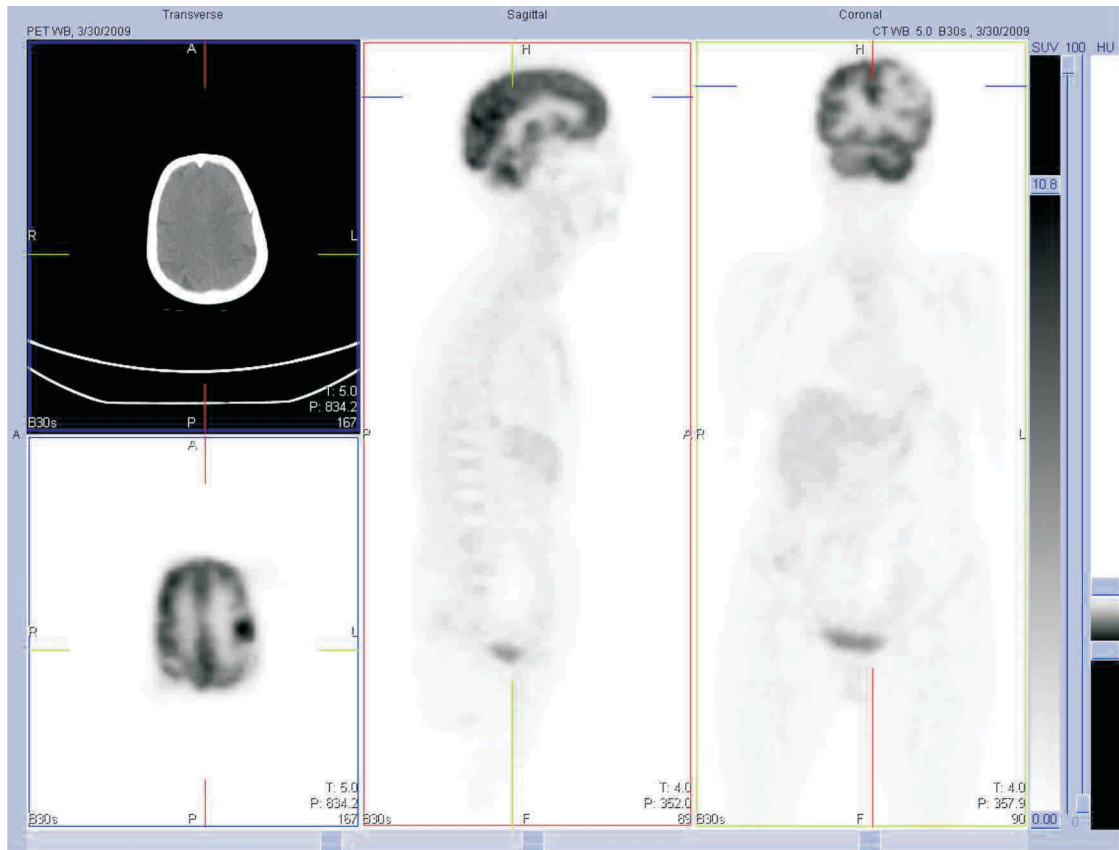


Fig 17 A PET scan showing left parietal tumor in axial section with evidence of crossed cerebellar diaschisis best seen in coronal section

Cerebral lesion:

A 72 years old lady had presented with weakness of right side of body for about one year. She became aphasic for 2 weeks prior to consultation. CT scan revealed gyral hyperdensity and effaced sulci in parietal cortex. A differential diagnosis of infarct, oligodendroglioma was made.

PET CT scan (Fig 17A) revealed a metabolically active lesion measuring 2 x 2 cm in the left parietal cortex with an SUV of 12.73. The surrounding brain parenchyma was hypometabolic which was probably related to oedema.

The contralateral cerebellar hemisphere revealed generalized hypometabolism- this was typical of crossed cerebellar diaschisis.

The cerebral lesion was neoplastic in appearance; however histologic evaluation could not be done.

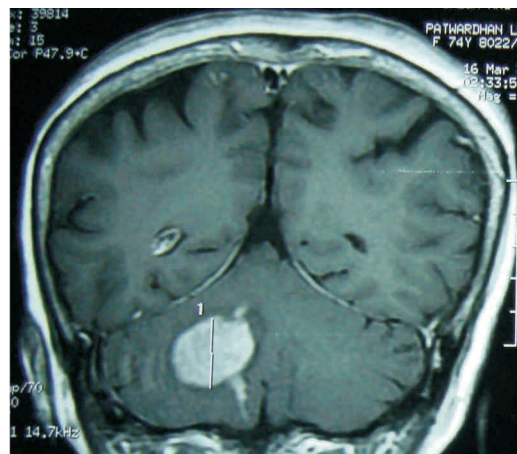
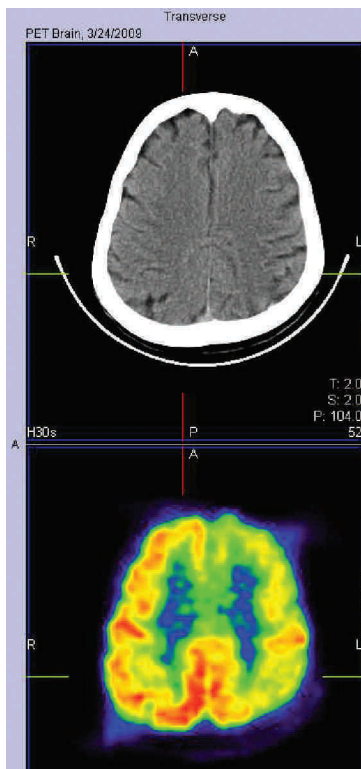


Fig 17 B MRI showing right cerebellar lesion

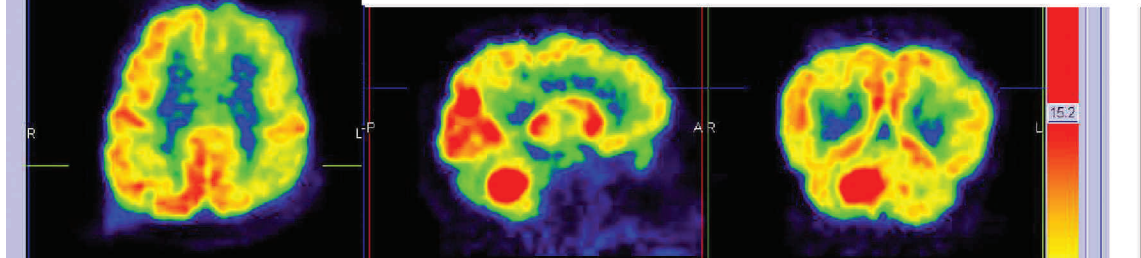


FIG 17 C PET Scan showing Right Cerebellar lesion that is metabolically active (Coronal & Sagittal sections) with evidence of hypometabolism in the opposite Cerebral hemisphere (Axial section)

Cerebellar lesion:

This 74 years old lady presented with imbalance of gait of one week duration. An MRI (Fig 17B) revealed right cerebellar enhancing lesion. A differential diagnosis of metastasis and granuloma was made. PET CT was ordered to look for any satellite lesions as there was an old history of nephrectomy for renal cell carcinoma.

PET CT (Fig 17C) revealed metabolically active right cerebellar lesion measuring 2 x 2.5cm with an SUV of 25. There were no satellite lesions within the brain.

The opposite frontal cortex revealed hypometabolism indicating crossed hemispheric diaschisis. As opposed to crossed cerebellar diaschisis, it was observed that lesions in the cerebellum account for hypometabolism in the opposite cerebral hemisphere. This is referred to as crossed hemispheric diaschisis. (Fig 17C)

Histology of the lesion revealed Lymphoma.

The term diaschisis was coined by von Monakow in 1914 from the Greek term meaning “shocked throughout” It implies remote functional changes. There is depression of regional neuronal metabolism and cerebral blood flow caused by anatomically separate but functionally related neuronal region. This occurs because of diminished excitatory trans-synaptic neuronal input into otherwise normal neurologic tissue with subsequent decrease in metabolic rate and alteration in blood flow.

Permanent menace :

Shree Narayanmurthy was imprisoned for 70 hours in a cell when he was found to have expressed his counter views on communism to a fellow passenger while traveling by train. Rosa Parks was imprisoned for thirteen months for not vacating her seat to a standing white passenger. She had violated a law which she rightfully thought was unjust. Eventually they had to change the law.

Are we not lucky? Freedom of expression is one right which has not been questioned in our country and is thankfully respected cross board.

“Permanent” a term so close to our hearts, a permanent job, a permanent house to live in, a permanent club membership, a permanent source of side income, a permanent vehicle etc. Apply it to a public sector, a permanently absent babu, a persistently negligent clerk, a perpetual reason for denying permission, a perpetual negative attitude, a recurrent chai break, a standard reason for missing line(lakir ke fakir) a permanently rude bus conductor, an equally rash bus driver, the list never ends.

Go to the private sector, a persistently harassed employer, an endless list of government permissions, a perpetually dissatisfied greedy babu, a never ending list of taxes, an equally long list of compliances, the enormously empowered inspector and the list never ends.

Now let's look into the statistics of any state. Take for instance; the state of Maharashtra which has about 17 lakh government employees. The total population is roughly about 9 crores. Approximately 2% of the people are responsible for so called governance and the remaining 98% are the “governed”. They are the one who I refer to as tax payers. (Even a beggar is a tax payer in one way or the other as there is no commodity that comes free of tax for making a livelihood)

Is there is deficit of educated unemployed individuals in this country? If not, then why the permanent government job benefits be given to just 2% of the population? Suppose the people decide that each government job will come with validity period of 5 years (just as a political post) the number of beneficiaries could easily encompass a large majority of people. It could then truly represent a democracy governed by the people for the people unlike the present state. There could of course be limited exceptions in the field of defense, education and judiciary.

(For those who don't know, I was once a government employee who could have easily enjoyed the “permanent benefits”)

“Unless a government servant is forced to make a living outside his protected environment for a major part of his life, he will not change his attitude ;

Unless a common man is given an opportunity to work as a government servant, he will not understand the importance of paying taxes.”

Address :

SPECT LAB

NUCLEAR MEDICINE SERVICES

K-2/1, Erandawana Co-op.Society,Opp. Dinanath Mangeshkar Hospital,Near Mehendale Garage,

Pune - 411 004.(INDIA).Phones: (020)25441125 / 25443596 / 25442557 / 25441128

Visit us at : www.spectlab.com Email : solav@vsnl.com